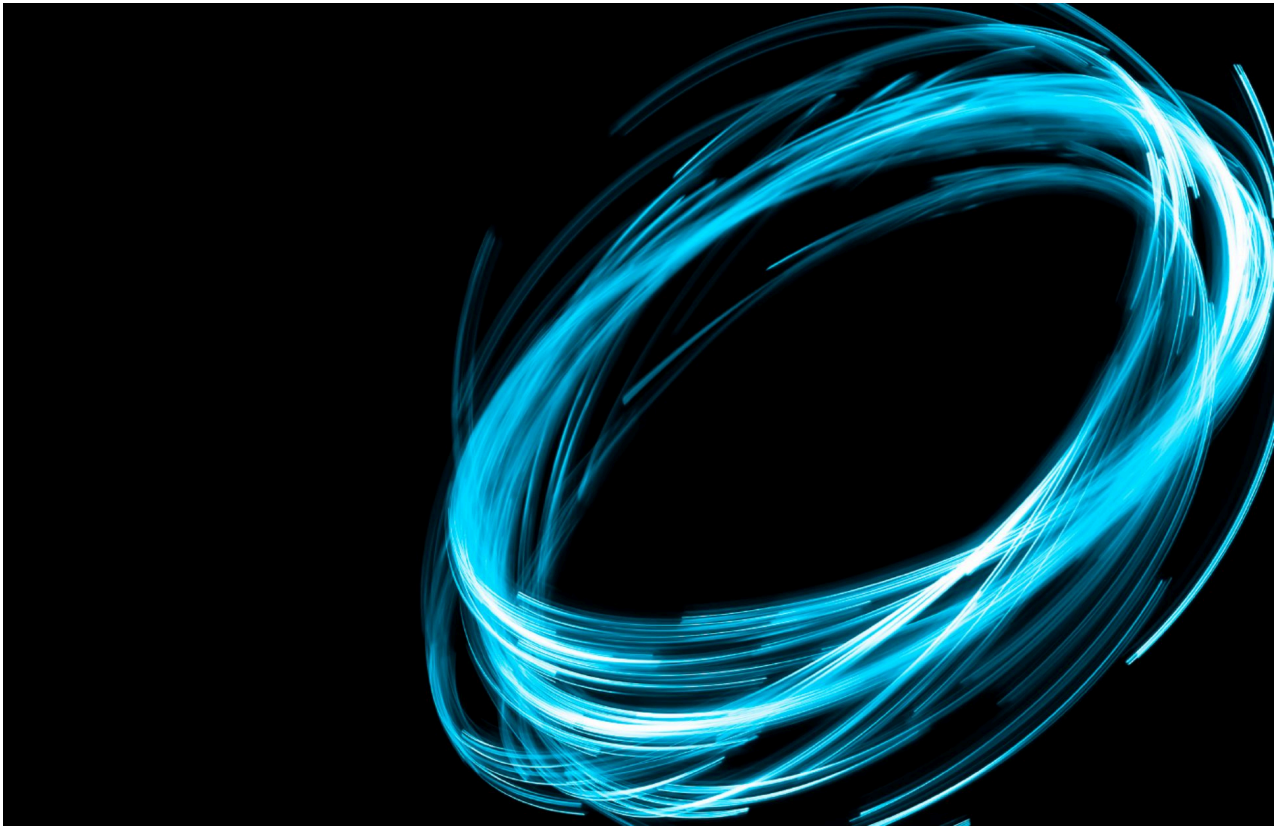


01/09/2023

Using natural language processing and machine learning in reg-tech applications



Muqiu Liu and Estelle Xu, Delta Analytics, and Chengcheng Wang, China Universal Asset Management, preview their IFoA Asia Conference 2023 presentation, which aims to empower professionals to navigate the complexities of the Asian market

As we step into autumn, a season of harvesting, following the challenging period of the COVID-19 pandemic, it's a privilege to be at this conference to have lively conversations with the elites.

The IFoA stands as a beacon of excellence in the actuarial and insurance industry, embodying unparalleled significance and influence. It serves as a transformative platform that shapes the trajectory of these fields through its steadfast commitment to advancing expertise and fostering global connections.

As the IFoA's flagship event in Asia, the conference holds a unique position in propelling the development of actuarial practices across various domains. Its multidisciplinary approach promotes cross-sectoral learning, reflecting the interconnected nature of modern challenges in the industry. As a hub of forward-looking ideas, the IFoA Asia Conference facilitates the exchange of such transformative knowledge, ultimately driving the actuarial and insurance sectors towards a more dynamic and promising future.

What is our topic about?

Our presentation topic, centred on machine learning and natural language processing techniques, resonates profoundly with the Asian financial market.

In this age of digital transformation, where data is the currency of decision-making, our exploration of these cutting-edge technologies aligns with the industry's evolving demands.

The Asian financial landscape is witnessing rapid innovation and disruption. Our insights into applying machine learning and natural language processing to this context offer invaluable solutions to extract actionable intelligence from vast data sets. By delving into the intricacies of AI-powered analysis, we seek to empower professionals to navigate the complexities of the Asian market, enhancing risk management, precision in decision-making, and overall competitiveness.

We've developed a machine learning-based quantitative model to identify key commonalities among target companies. Qualitative investigations help determine criteria for effectively screening potential candidates. Our customised model incorporates selected indicators and target companies to present determinants and concise representations of their shared characteristics, enabling systematic ranking and descriptions for newly added companies.

To harness unstructured data, we've integrated natural language processing techniques to process textual content from companies' disclosures, identifying key entities and financial terms through named entity recognition. This

informed investment decisions.

As an example, we used natural language processing to assess a company's innovative ability. Prospectuses and annual reports are automatically read, key indicators extracted, and a holistic comparison among peers conducted. This integrated methodology is poised to enhance credit ratings, ESG screening, audits, and other applications, offering comprehensive investment insights.

We believe every small pace of trying aggregates a giant leap of advancement. Our presentation and paper merely scratch the surface of these two technological applications. However, we hope that through this sharing, participants will unleash their imagination and collectively drive progress in the realms of finance and technology.

Learn more at the IFoA Asia Conference 2023

Don't miss Muqiu Liu, Estelle Xu, and Chengcheng Wang at the IFoA Asia Conference 2023 on 26 September, in Kuala Lumpur. [Book your place before 15 September.](#)



Authors



[Muqiu Liu](#)

Delta Analytics



[Estelle Xu](#)

Delta Analytics



[Chengcheng Wang](#)

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